GAP-10T Split-Jet Right Inverse: Exact Generic-GR Kinematics with a Nonpropagating Composite Jet Connection

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Abstract

The minimal one-connection architecture cannot simultaneously provide universal single-Θ representability and exact generic torsion-free GR. This note constructs the smallest local kinematic completion after that no-go. Physical curvature and matter transport use the Levi–Civita connection of an arbitrary prescribed tetrad. Only the defining Θ jet is augmented by an explicitly composite Lorentz contortion and a composite real dilation one-form. For every non-null Lorentz-real field X and every smooth tetrad E, the formulas below give $\widehat{D}_\mu X = \sqrt{\mathcal{N}_0} E_\mu$ identically. The added jet objects are algebraic in $(E, X, \mathring{D}X)$ and introduce no independent propagating field. The construction closes local exact-GR representability of the split-connection architecture. It does not derive the tetrad from Θ, select the jet representative from an action, or derive the Hilbert–Palatini coefficient.

1 Physical and jet connections

Let $E_\mu = e_\mu{}^a \mathbf{u}_a$ be an arbitrary smooth nondegenerate Lorentz tetrad and let $\mathring{\omega}(E)$ be its Levi–Civita spin connection. This is the *physical* connection:

$$\Omega_{\text{phys}} = \mathring{\Omega}(E), \qquad T_{\text{phys}} = 0, \qquad R_{\text{phys}} = R[\mathring{\Omega}(E)]. \tag{1}$$

Therefore R_{phys} is as general as the Riemann curvature of the prescribed GR tetrad.

Let $X = X^a \mathbf{u}_a$ be the Lorentz-real projection of Θ on a patch where

$$X^2 := \eta_{ab} X^a X^b \neq 0, \qquad s := \sqrt{\mathcal{N}_0}. \tag{2}$$

Define the covariant mismatch

$$Z_\mu{}^a := s e_\mu{}^a - \mathring{D}_\mu X^a. \tag{3}$$

Decompose it into its component parallel to X and its orthogonal component:

$$w_\mu := \frac{X_a Z_\mu{}^a}{X^2}, \qquad Z_{\mu\perp}{}^a := Z_\mu{}^a - w_\mu X^a, \qquad X_a Z_{\mu\perp}{}^a = 0. \tag{4}$$

2 Exact algebraic right inverse

Theorem 1 (Split-jet right inverse)**.** *Define the Lorentz-algebra-valued composite tensor*

$$\boxed{K^J_{\mu ab} := \frac{Z_{\mu\perp a} X_b - X_a Z_{\mu\perp b}}{X^2}, \qquad K^J_{\mu ab} = -K^J_{\mu ba}.} \tag{5}$$

Let the jet derivative on X be

$$\widehat{D}_\mu X^a := \mathring{D}_\mu X^a + K^{J\,a}{}_{\mu b} X^b + w_\mu X^a. \tag{6}$$

Then

$$\boxed{\widehat{D}_\mu X^a = s e_\mu{}^a} \tag{7}$$

holds identically for every (E, X) on the non-null patch.

Proof. Equation (4) makes $Z_{\mu\perp}$ orthogonal to X. Contracting Eq. (5) with X^b gives

$$K^{J\,a}{}_{\mu b} X^b = Z_{\mu\perp}{}^a. \tag{8}$$

Thus

$$\widehat{D}_\mu X^a = \mathring{D}_\mu X^a + Z_{\mu\perp}{}^a + w_\mu X^a = \mathring{D}_\mu X^a + Z_\mu{}^a = s e_\mu{}^a. \tag{9}$$

□

The general Lorentz solution differs from Eq. (5) by a stabilizer term $C_{\mu ab}$ satisfying $C^a{}_{\mu b} X^b = 0$. Such a term is invisible in $\widehat{D}X$; Eq. (5) is the covariant representative with no stabilizer component.

3 Biquaternionic bimodule form

Let $\mathcal{K}^J_\mu$ be the spin lift of $K^J_{\mu ab}$. The real one-form w_μ can be implemented by a relative central action:

$$A^J_\mu = \mathring{\Omega}_\mu + \mathcal{K}^J_\mu + \frac{1}{2} w_\mu \mathbf{1}, \qquad B^J_\mu = -\mathring{\Omega}^\ddagger_\mu - \mathcal{K}^{J\ddagger}_\mu - \frac{1}{2} w_\mu \mathbf{1}. \tag{10}$$

Then $\partial_\mu X + A^J_\mu X - X B^J_\mu$ is precisely Eq. (6). The relative central term is not a second physical gauge field: it is the fixed composite scalar in Eq. (4) and acts only in the defining jet. It preserves the Lorentz-real slice but is not required to preserve the Lorentz norm under jet transport. Physical metric compatibility remains entirely with Eq. (1).

Corollary 2 (Generic GR curvature is kinematically available)**.** *For every local GR tetrad, including Schwarzschild, Kerr, FRW, and generic numerical-relativity tetrads, choose any smooth non-null local X. The split-jet construction represents that tetrad while the physical curvature is exactly its Levi–Civita curvature. No concurrent-vector restriction is imposed on Ω_{phys} because the concurrent derivative is $\widehat{D}$, not the physical Levi–Civita derivative.*

4 Boundary of the result

The construction is a local right inverse, not a dynamical map $\Theta \mapsto E$. Because K^J and w are defined using the prescribed E, Eq. (7) does not select one tetrad among all possible tetrads. A complete UBT derivation still needs an action or constraint principle that:

1. selects the split jet structure without turning K^J or w into a propagating physical field;
2. supplies Einstein curvature dynamics with the correct sign and Newton coefficient;
3. gives a well-posed degree-of-freedom count and on-shell solutions.

The construction also requires patching through $X^2 = 0$ or using complementary non-null representatives.

Exact status

GAP-10T-JET-KIN: CLOSED LOCALLY [L1]. The split architecture has an explicit local composite/nonpropagating jet right inverse for every tetrad and every non-null Lorentz-real X, while physical curvature remains exactly Levi–Civita and therefore as general as GR.

GAP-10T-JET-AUX: CLOSED [L1]. The follow-up multiplier action in `gap_10t_split_jet_auxiliary_completion.tex` proves that the jet variables are algebraic, nonpropagating, and on-shell decoupled on every non-null patch.

GAP-10T-JET-CONSTRAINT-SELECTION: CLOSED AS NO-GO [L1]. The surjective pure constraint cannot select a unique physical tetrad.

GAP-10T-JET-DYN: NARROWED. The canonical action does not yet select the physical tetrad and representative or derive a separate non-surjective metric effective law; null-patch/global continuation and the constrained mode measure remain.

GAP-10D: NARROWED. The Einstein endpoint is structurally available, but the curvature action and Newton coefficient are not derived from the original canonical UBT dynamics.